Ferrules for boiler tubes should be soft copper, 1/32 inch longer than the thickness of the back tube sheet.

Ferrules must be of the same outside diameter as the tube and should be 0.0625 (1/16 inch), 0.08 and 0.09 inch thick. For new work, ferrules should be 1/16 inch thick, the heavier copper being used after the tube holes have become enlarged through service. This practice permits the use of a standard swage. Copper ferrules may be softened by annealing.

Reaming Tube Holes—When a firebox sheet tube hole becomes ⅛ inch out of round, ream it with a straight

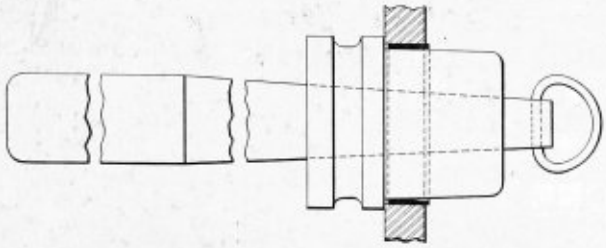


Fig. 2.—Setting Copper Ferrule in Tube Sheet

reamer. When it becomes enlarged through service to 3/16 inch above outside diameter of tube, scrap the sheet.

Preparation of Firebox End of Tubes—Swage the firebox end of the tube with a slight taper toward the end of the tube to ⅛ inch less than the outside diameter of the tube, so as to drive snugly in the ferrule after the ferrule has been tightened in the sheet. Length of swaged end should be the thickness of the tube sheet plus ¼ inch.

Preparation of Front End of Tubes—Cut tubes to length. The front ends should be expanded hot, to such a size that when placed in the boiler they can be driven snug in the front tube sheet. This should not be done to tubes that are to be transferred (put through one hole and set in a different location) in the boiler.

The ends of the two tubes that are to be placed in the enlarged holes may either be expanded hot to the required diameter or have a piece of tube of larger size, 12 inches long welded on.

Setting Ferrules—Place ferrules in the tube holes in the firebox sheet, 1/32 inch from the fire side. Tighten with

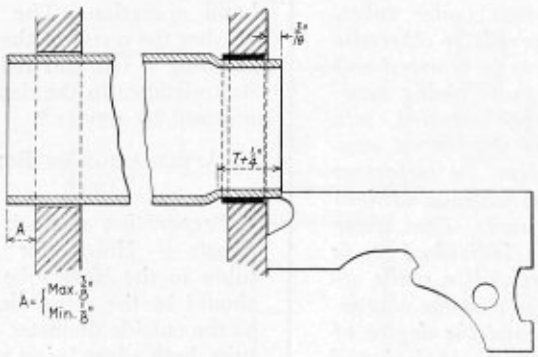


Fig. 3.—Proper Position of Tube in Sheet

straight sectional expander, Fig. 2, or roll with roller expander.

Placing Tubes—Clean the firebox ends of the tubes and place them in the tube holes in the firebox sheet, fitting them neatly into the copper ferrules.

Be careful not to injure or displace the ferrules.

Tubes should project through the firebox sheet 3/16 inch, as measured by the gage, Fig. 3. The projection through the front tube sheet should be ¼ inch. It must not be less than ⅛ inch nor more than ⅜ inch.

Tightening Tube in Sheet—Tighten tubes in the firebox sheet with a straight sectional expander, Fig. 4.

Tightening Front End of Tubes—Next tighten the front end of the tube with a standard roller expander.

Flaring Tubes, Preparatory to Beading—The firebox end of all tubes, and on superheater engines the front ends of the two rows of tubes just below the superheater flues and all tubes above these two rows, should be opened with a flaring tool, Fig. 6. Use a long stroke pneumatic hammer.

Expanding with Lipped Sectional Expander—Expand the firebox ends of the tubes with the lipped sectional expander shown in Fig. 7. Drive the expander pin lightly until it is fairly solid. Draw out the pin. Give the expander a turn equal to half a section of the expander. Drive the pin a second time. Withdraw it, and turn the expander a distance equal to one-quarter of a section. Drive the pin a third time. Care should be exercised in the use of the expander in the outer rows of tubes near the heel of the flange to avoid cracking the sheet. Use a long stroke Boyer No. 60 pneumatic hammer (a heavier hammer is not recommended) in driving the pin or its equivalent.

Inspection for Split Tubes—After expanding, examine the tubes carefully. Those which have split in the shoulder formed by the lip of the expander, and tubes that have opened from the end back to the sheet, should be removed and replaced before beading.

Opening Front End of Tubes—If the front end of the tube has not been opened hot before placing in the boiler,

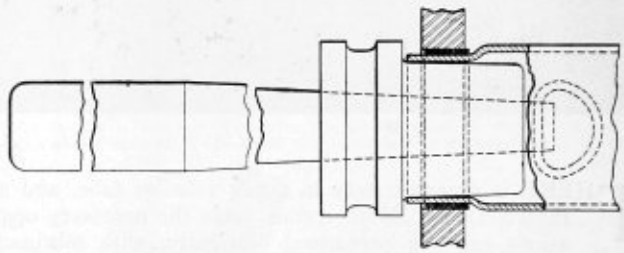


Fig. 4.—Method of Tightening Tube in Back Tube Sheet

a straight sectional expander should be used to tighten the tube in the sheet. The tube may then be rolled with a standard roller expander.

If the finger type roller expander is used, the use of the straight sectional expander is not required.

Beading—Firebox ends of all tubes must be beaded, Fig 9. Use a short stroke pneumatic hammer. Take care that nothing enters between the bead and the tube sheet. In superheater locomotives the front ends of the two rows of tubes just below the lower row of superheater flues and all tubes above these rows must be beaded similarly.

Hold the beading tool so as to give the bead an outward set and get it down to the sheet without raising a burr on the inside or marking the outside of the sheet with the heel of the beading tool. The center line of the beading tool should always be inside the line of the tube. Remove any burrs that may be raised with a small chisel and hand hammer. If necessary, the firebox ends of the tubes may be rolled lightly after beading.

Tools—All beading tools must be maintained within the limits of standard gages. Lipped expanders should be stamped to show the thickness of sheet on which they are to be used and should be taken out of service when worn or distorted to such an extent that they will not force the tube squarely against the sheet.

Application of Superheater Flues

Preparation of Back Tube Sheet—Holes for superheater flues in the back tube sheet should be 4⅝ inches, both edges being rounded to a radius of 1/16 inch.

Preparation of Front Tube Sheet—Holes in the front tube sheet should be 3/32 inch larger than the outside diameter of the flue and the outside edges should be rounded to a radius of 1/16 inch.

Copper Ferrules—Ferrules must be made of soft copper, 1/32 inch longer than the thickness of the tube sheet. Ferrules must be of the same outside diameter as the holes in

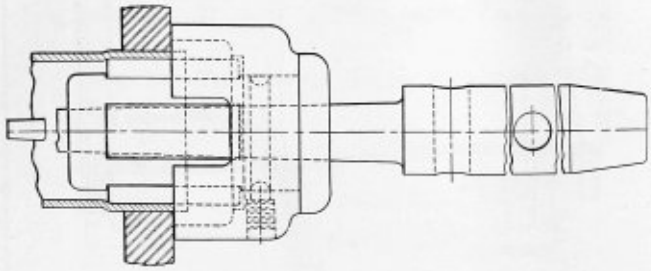


Fig. 5.—Rolling the Tube in the Front Tube Sheet

the tube sheet and should be 0.0625, 0.08 and 0.09 inch thick. For new work, use ferrules 1/16 inch thick, the heavier copper being used after the holes become enlarged through service.

Preparation of Firebox Ends of Flues—Firebox ends of flues should be swaged to a diameter of 4½ inches. The straight portion should be 5 inches and the tapered portion 3 inches long, as in Fig. 10. Remove all scale and burrs before placing the ends in the tube sheet.

Preparation of Front End of Flues—Cut flue to length. Expand front end, hot, to a diameter 1/16 inch greater than the outside diameter of the flue, as in Fig. 10.

Setting Ferrule—Place the ferrules in the flue holes in the back tube sheet 1/32 inch from the fire side and tighten with the roller expander, Fig. 11.

Placing Flue—Place flues in the holes in the back-tube sheet, fitting them neatly into the copper ferrules. In placing flues, be careful not to displace copper ferrules. Flues should project through both sheets ¼ inch to allow for beading, Fig. 10.

Rolling Flue in Back Tube Sheet—Tighten flues in the

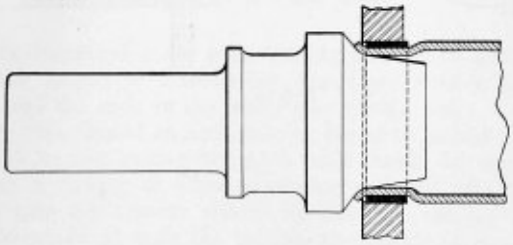


Fig. 6.—Method of Flaring Tubes

back tube sheet with a roller expander. The roller expander shall have not less than 5 rollers, Fig. 12.

Flaring Flues in Back Tube Sheet—A flaring tool, because of its size and weight and the force necessary to drive it, is not recommended for superheater flues. Superheater flues should be flared by a four-pound peining hammer.

Expanding Flues in Back Tube Sheet—Expand firebox ends of flues with lipped sectional expander of not less than 12 segments, Fig. 13.

Drive the expander pin until it is fairly solid. Draw out the pin and give the expander a turn equal to one-half of a section of the expander. Drive the pin a second time. Withdraw it. Turn expander a distance equal to one-quarter of a section. Drive the pin a third time. Use a long stroke pneumatic hammer in driving the pin.

Rolling Front Ends of Flues—Tighten the front ends of flues with a roller expander, Fig. 14.

Flaring Front Ends of Flues—A flaring tool, because of its size and weight and the force necessary to drive it, is not recommended for front ends of flues. Front ends of flues should be flared by careful blows of a four-pound peining hammer.

Beading Superheater Flues—Bead the flues at both ends, Fig. 15, using a short stroke pneumatic hammer. Take care to see that nothing enters between the bead and the sheet. Hold the beading tool so as to give the bead an outward set and get it down to the sheet without raising a burr on the inside, or marking the sheet with the heel of the tool on the outside. The center line of the beading tool should al-

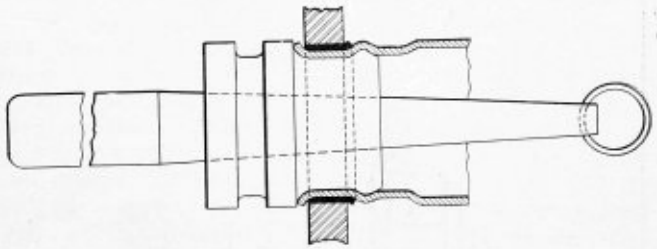


Fig. 7.—Lipped Sectional Expander Used to Expand the Firebox End of Tubes

ways be inside the line of the flue. Remove any burrs that may be raised with a small chisel and hand hammer.

Tools—All beading tools must be maintained within the limits of standard gages. Lipped expanders should be stamped to show the thickness of sheet on which they are to be used and should be taken out of service when worn or distorted to such an extent that they will not force the tube squarely against the sheet.

APPLICATION OF ARCH TUBES

Preparation of Sheet—Sheets must be properly straightened. Holes for arch tubes must be circular and drilled 1/32 inch larger than the outside diameter of the arch tube. Outside edges should be rounded to a radius of 1/16 inch.

If the arch tube hole has become enlarged above normal size in service, measure it carefully, so that the tube can be heated and enlarged to fit the hole snugly before being applied. If this is not possible a copper ferrule may be used extending ¼ inch on the water side and flush with fire side of the sheet.

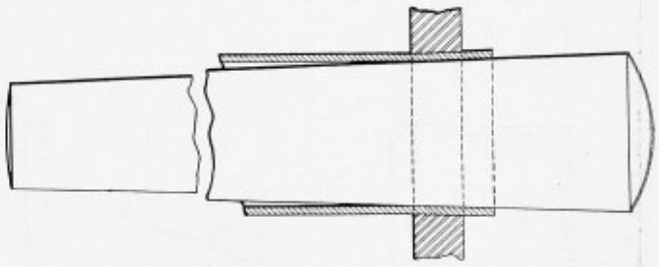


Fig. 8.—Pinning Out the Front End of a Tube

Preparation of Arch Tubes for Setting—Cut arch tubes to proper length, with an allowance of ½ inch at either end for flaring. They should be smooth at the ends and free from burrs. They should be formed in accordance with drawings for the various classes of locomotives. Particular attention should be given to bending, in order to prevent the formation of steam pockets in service due to improper bends to make certain that the arch tubes enter the sheets at an angle of

which it was naturally subjected and the added stress thrown upon it due to the six broken staybolts.

APPEARANCE OF FIREBOX

The interior of the firebox indicated that all the sheets had been overheated, due to the fact that the fire was not drawn until 15 or 20 minutes after the boiler had emptied itself. The door sheet and flue sheet seams were slightly sprung, due to the intense heat; but no damage of a serious

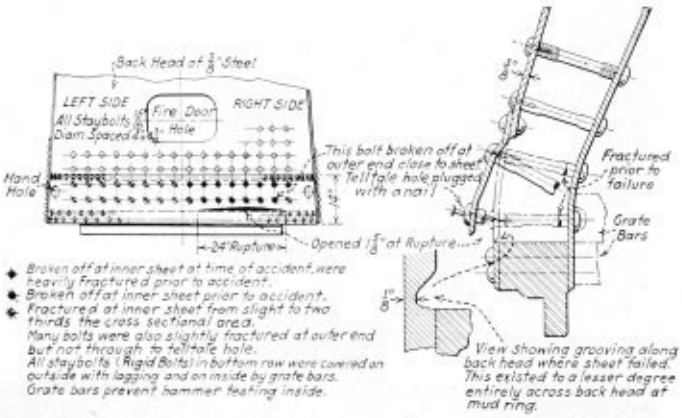


Fig. 2.—Diagram Showing the Location and Causes of Failure

nature was caused by the overheating. The general condition of the firebox, including flues and arch tubes, was good.

Unless removed, the grate bars completely prevented the hammer testing of the entire bottom row of staybolts from the inside of the firebox and the lagging and jacket covered the outer end of the bottom row of bolts in the back head.

After observing the failed portion of the sheet it was cut out in such a manner as to reveal the true condition of the staybolts and so as to determine the extent of the corrosion across the back head close to the mud ring. This disclosed the dangerous condition previously described as regards the defective staybolts and the same destructive grooving all the way across the back head close to the mud ring.

To ascertain if the steam pressure carried was in excess of the allowed working pressure at the time of the accident, the steam gage, after being tested on a deadweight tester and being found approximately correct, together with the safety valves from the defective locomotive, were applied to Pennsylvania Railroad locomotive 2250 of the same class and the pressure raised until the safety valves opened at 205 and 209 pounds, respectively, according to the tested gage, which removes all question as to overpressure.

The company's records indicate that the locomotive was built by the Baldwin company during March, 1905. A firebox was applied at Altoona, Pa., in May, 1916, and class 3 repairs given on September 5, 1918, and class 5 on May 16, 1920. There was no record obtainable covering application of the back head patch. The last annual inspection was made on April 20, 1921; the last monthly inspection was made at Shire Oaks, Pa., on November 7, 1921, at which time the staybolts were shown as "good" and none applied.

DAILY INSPECTIONS MADE

Daily locomotive inspection reports were carefully examined at Shire Oaks and Monongahela City, Pa., for a period of 15 days prior to the accident and nothing was found bearing on the accident except—

On December 4, engineer F. G. Jarrett reported at Monongahela City, Pa., at 4 p. m.:

"Mud seam leaking bad."

The locomotive was afterwards run to the shop at Shire Oaks, a few miles from Monongahela City, for repairs. Upon arrival there, it was found that the bad leak reported came from a telltale hole in a staybolt located in the bottom row on back head near the right back corner, and that the mud seam or ring was not leaking. The foreman in charge ordered the staybolt plugged. This plugging was done sometime between 7 p. m. and midnight of December 4 and the locomotive returned to service at 2:15 a. m. on December 5 with the staybolt plugged.

Engineer Jarrett made the next report, but no mention was made of the leak, which indicates that the leak reported the previous trip was from the telltale hole in the broken staybolt under the jacket instead of the mud ring. Engineer Jarrett verified this during the investigation.

Had it not been for the broken staybolts referred to, the worst that would have resulted from the grooved sheet would have been a leak, even if the grooving had extended entirely through.

The condition of these staybolts should have been disclosed and this fatal accident averted had they been tested and repairs made as required by the rules.

To have properly complied with the rules it would have been necessary at each inspection to have removed the grate bars which covered the inner end of the first row of staybolts

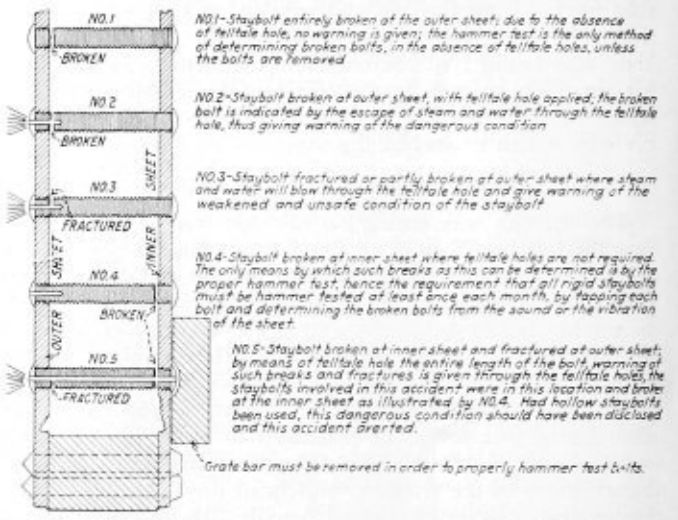


Fig. 3.—Types of Staybolt Fractures with Telltale Holes as Indicators of Breaks

above the mud ring, or to have complied with the rules in the most technical sense, it would have been necessary to have removed the jacket and lagging from the back head which covered the outer end of the bolts. Evidence indicates that it has not been the practice of this carrier to make a proper hammer test of staybolts located behind grate bars nor to apply staybolts with telltale holes their entire length.

REQUEST FOR STAYBOLTS WITH TELLTALE HOLES

At the urgent request of a number of railroads for permission to apply staybolts with telltale holes their entire length behind brickwork or oil-burning locomotives, to avoid the necessity for removing the brickwork each time the staybolts are tested, the chief inspector of this bureau issued, on November 11, 1913, Circular 118, as follows:

If staybolts which are behind brickwork on oil-burning locomotives or behind grate bearers have a telltale hole three-sixteenths inch in diameter their entire length which is kept open at all times, the removal of the brickwork or grate bearers each month for the purpose of hammer testing such bolts will not be required. This will not, however, relieve from making a thorough inspection each time the brickwork is removed nor will it relieve from removing the brickwork for an inspection when necessary.

So far as it has been brought to our knowledge the railroads have taken advantage of this circular and applied

boiler with water, sand blasting and then going over the flues with the oxy-acetylene torch, in order to heat them up enough to get any grease out where the welding will take place.

John Harthill, Cleveland, New York Central R.R.: We use absolutely no grease in preparing the back flue sheet; in drilling the flue holes, we use a soda compound. We apply our flues just the same with the copper ferrule and take just as much pains to work the flues as though they were not to be welded.

In setting the flues, we use soft soap for lubrication of all tools. In doing this, when you apply heat to the flue the soap dries and falls off. If oil were used it would run around the flue to the end of the weld, and there is where you later have a leak. After the flue is welded, we use a beading tool and that roughs the flue and brings up the bright metal; then we weld the flue.

I cannot see why you should weld flues in a boiler with water on one side and the extreme welding heat on the other.

J. A. Anderson, Industrial Works, Bay City, Mich.: Our boilers are the vertical type, for the most part, and we are thinking about welding the beads of the tubes at the end. Any information on this work will be very much appreciated.

T. W. Lowe, C. P. R.R., Winnipeg, Man., Can.: We have been working on a plan of our own, and I want to tell you that we weld all our superheater flues with good success. We also have applied a new firebox to one engine with superheater flues and the smaller flues all welded to find out what we can do with them.

We had complete success until the time came that these small flues required to be worked upon. We ran that engine about forty thousand miles before we had to work those small flues and at that time we found it was very difficult to maintain them tight. It was undesirable to have them welded because they cannot be worked as well as when they are applied without welding.

G. W. Bennett, I. C. C., Albany, N. Y.: Several years ago, Mr. Lowe made a statement that he was applying tubes up in Canada in the bad water district without the copper ferrule. I would like to ask him if he is still following that practice.

T. W. Lowe, C. P. R.R.: We have returned to the copper ferrule in the superheater flues. We have always used the copper ferrule in the small flues.

C. A. Nicholson, Southern Railway, Atlanta: Years ago, we never got over nine months from flues. At the end of nine months, it was necessary to remove these flues. Now with welded flues we get four years and, in some cases, on our new engines the federal inspectors have granted us an additional year, and they would grant more, possibly, if we would ask.

We also have bad water conditions, but we have one division where we do not have to weld the flues at all, they will run four or five years. We have divisions going into Florida and Alabama where it is necessary to remove part of the flues in a year, even now with the welding, but during the months that these flues are in the boiler, we have cut the roundhouse forces practically in two.

We have tried several ways of welding flues and after going through them all, I find that the best way of application of the weld is after the engine has been fired up or run a week or even two weeks, but it should be as early as that in order to get good service out of them.

We weld them without water and with water. We clean them with a beading tool, we do not use the sand blast; the better they are cleaned, the better they will hold.

W. P. Harrison, A. T. & S. F. R.R., La Junta, Colo.: Relative to vertical stationary boilers, I would like to say that we have one at Mohanta, Colorado, located on a clam shell. We had a considerable lot of trouble with flues leaking, in fact, we renewed the flues as often as once each six months.

About a year ago, we renewed those flues and beaded them and expanded them on both ends and electric welded them to the sheet on both ends. Since that time, they have never been worked any more. The clam shell is now in the shop for repairs, but there will be nothing done to the flues because they are in first class condition; that trouble was overcome by welding in the flues.

A. N. Lucas: I would like to ask the gentleman who just spoke, if electric welding flues in this vertical boiler stopped the gathering of scale on top of the flue sheets. If it didn't, your flues are going to leak. I have had experience with vertical type boilers, and I never had any trouble with them leaking until the mud got there, then I would pull a few flues and go ahead again. Then we changed our plan and got the mud off, then we got service again, but on vertical type boiler welding the flues will not help unless you keep your flue sheet clean.

W. P. Harrison: I would like to say we worked the boiler before we welded the flues and since, but the flues have not leaked for one year. Up until that time, they were continually leaking. We are using the same water and under the same conditions, therefore, I am bound to say that welding in the flues was what gave us the results which we are now having.

H. B. Stevens, Santa Fe, Topeka: I believe the electric welded flue is a success, and they should be applied in the usual manner of applying flues, and carefully beaded down, before the engine goes into service, it should be fired up and give the flues a chance to set and the grease burned off, then weld with the proper welding fire and not get too much



Frank Gray, Second Vice-President



T. F. Powers, Third Vice-President



J. F. Raps, Fourth Vice-President

in suburban service. These locomotives operate in both passenger and freight service on mountainous and rolling territory. Both coal and oil are used for fuel.

One of the most important considerations in the selection of the type of feed water heater to be used, is the character of the water in the territory through which the locomotive is to operate. In bad water districts, the open type heater seems to be preferred, as the scale deposits on the tubes of the closed type heater retard the heat transmitted and reduce the efficiency of the heater, and there would be less danger of boiler trouble from oil due to the frequent washouts. No road has reported trouble from oil from the feed water in the boiler. Roads where the boiler washout period averages 30 days generally prefer the closed type of feed water heater.

Three roads have reduced the size of the exhaust nozzles on application of feed water heaters and one road has enlarged the nozzle. The reduction in the size of the nozzle is done in order to offset the loss in superheat temperature which occurs when a feed water heater is applied to a locomotive. This is not considered advisable, as the reduction of the size of the nozzle increases the back pressure, which will probably offset any saving that would be effected by the increased superheat.

There has been no difference reported in the amount of boiler scale in boilers equipped with feed water heaters over the other engines.

The open type heater has in all cases gone from shopping to shopping without cleaning, regardless of the water conditions. While going through the shop the scale deposit is scraped from the inside of the heater, no acid or cleaning solution being used.

In good water territory, the closed type of heater is cleaned each time the locomotive is shopped. The usual method of doing this is to dip the tube nest into a lye vat to remove any oily deposit which may have formed on the outside of the tubes. In districts where the water conditions necessitate more frequent cleaning, the deposit is either washed out by flushing the heater with water or, if the scale is not soluble, a dilute solution of muriatic acid is pumped through the heater for a short time and then water is pumped through to clean out the acid. The strength of the acid varies from 20 to 33⅓ percent.

The highest cost of cleaning the feed water heater is $170 per year, both labor and material, and the lowest cost is $2.31. In one case the heater is cleaned by the use of dilute muriatic acid; the other, by a basic solution. An average of the cost data submitted by all the roads for cleaning by the acid progress is $62.19 per heater per year, which includes both labor and material.

The cost of other maintenance of the heater proper per year is practically nothing on both the open and closed types. Where weak acid solutions are used, none of the heaters cleaned show any signs of deterioration due to the use of the acid. The territory in which locomotives equipped with feed water heaters operate, includes the greatest range of climatic conditions possible. No serious difficulty has been encountered with any of the feed water heater systems freezing up. Drain valves and telltale pipes have frozen up, but these have given no further trouble after lagging.

Failures of the heater proper while in service have occurred, due to tubes bursting or becoming loose in the tube sheet, heater heads cracking with the closed type of heater, and a crack developed in the cylinder near the discharge valve on the open type of heater. The brass tubes in the closed heater have been replaced by copper tubes, which are more ductile than the brass and a better joint can be made when the tubes fasten into the header. Some trouble has been experienced with the boiler checks pounding out or breaking off with the use of feed water heater equipment. This has been largely overcome by using larger boiler checks with reduced lift.

The boiler feed pump has given good service with all types of heaters. The most common defects which have been encountered are the pump piston rod packing leaking, rods wearing, water valve springs breaking, water cylinder scoring, top head pump gasket leaking, abnormal lift of intake and outlet valves, and valves stuck or leaking. The average cost of maintenance per pump, taken from the data submitted, is $55.16 per year, including both labor and material.

The cost of maintenance of the feed water heater apparatus complete averages $97.15 per locomotive for labor and material per year. This figure was determined by averaging the maintenance costs submitted by all the roads, regardless of the type of heater used.

As all the locomotives, which are equipped with feed water heaters, have an injector, no engine failures could be attributed to the operation of the feed water heater apparatus, as the injector was used to supply the boiler in case of the failure of the feed water heater.

Where feed water heaters are applied, the enginemen should be personally instructed relative to the operation of the equipment in order that the highest efficiency may be obtained. The feed water heater pump should only be used to supply the boiler when the engine is working steam, as the exhaust steam from the auxiliaries is not sufficient to heat the water hot enough to show a saving, and the introduction of the cold water into the boiler would tend to cause serious strains in the flues and flue sheets. The rule to pump locomotives with feed water heaters only when working steam, is in force on very nearly all the roads.

It has been possible to eliminate some water tank stops when the condensate from the feed water heater is returned to the boiler.

On tests made on feed water heater locomotives, the feed water heater shows a saving of between 10 and 16 percent in fuel, based on the evaporation performance of the locomotive. On two roads where dynamometer car tests had been conducted on feed water heater locomotives, there was a saving of approximately 10 percent based on fuel consumption per drawbar horsepower. On a thousand ton mile basis, the saving varies from 4 to 11 percent.

The accompanying power performance chart is based on data taken on dynamometer car tests. It shows that with the feed water heater locomotives, there is an increase of both indicated and drawbar horsepower. This increase in horsepower is due to the back pressure being decreased by diverting about 12 percent of the exhaust steam from the cylinders to the feed water heater, thus increasing the steaming capacity of the locomotive at high speeds. This increased horsepower will permit an increase in the tonnage rating and average speed of the locomotive.

In its present form the locomotive feed water heater has passed through the experimental stage in this country and the results indicated in this report are typical of what may be anticipated from the application of feed water heaters on other railroads, barring unusual local conditions.

At the present time, the exhaust steam injector as extensively used abroad is being considered as an alternative to the open type of locomotive feed water heater. One American firm is already engaged in the manufacture of this device, and arrangements are being perfected for supplying the railroads in this country with a type of exhaust steam injector that has been successfully used on an extensive scale in England and her colonies.

The report was signed by E. E. Chapman, chairman (A. T. & S. F.); J. R. Alexander (Penn.), E. A. Averill (Superheater Company), J. A. Carney (C. B. & Q.), J. N. Lammedee (Worthington Pump & Mchy. Corp.), L. P. Michael (C. & N. W.), Geo. E. Murray (Grand Trunk), C. B. Peck (Railway Age), L. G. Plant (Railway Review), G. B. Von Boden (Sou. Pac.), W. H. Winterrowd (Can. Pac.).

If the pitted or worn surface is not too great, it may be cut off and the good portion of the tube used in boilers having shorter tubes.

When tubes, otherwise good, have been reduced in weight below the limits given in Table 1, they should be discarded.

Tubes that are slightly pitted or otherwise weakened, but which are still serviceable, should be kept together and reapplied in sets of tubes which are in approximately the same condition.

Safe Ending

Good tubes should be safe ended as follows:

Safe ends should be scarfed on one end for half an inch, see Fig. 10, being tapered to 1/32 inch at end of scarfing.

Tubes to be safe ended should be expanded for the scarfed end of safe end so that they will fit snugly.

Place the safe end in end of the tube. Heat both to a welding heat. Draw out and quickly weld on the flue welding machine, taking care to see that the weld is smoothly made.

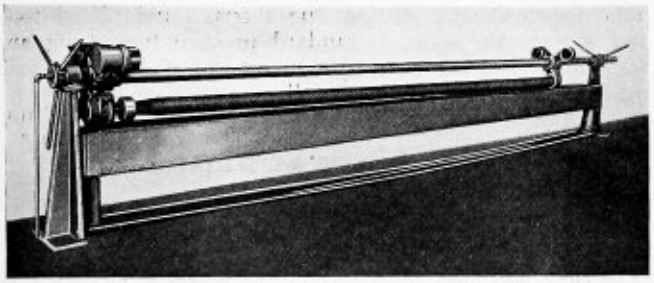


Fig. 9.—One Type of Tube Testing Machine

Successive Safe Ending

It is recommended that body tubes contain not more than three welds at one time, with a total length of welded ends not to exceed 21 inches. If the total lengths of welds exceed 21 inches, all welds should be cut off and one piece of the tube welded on to bring the length up to the requirement. To this a new 5-inch safe end should be added at the firebox end at the first safe ending. Thereafter Table 2 should govern.

Superheater flues should not contain more than two welds.

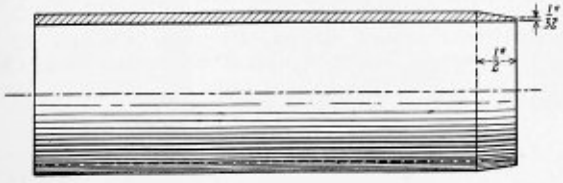


Fig. 10.—Correct Method of Preparing Safe Ends for Use

The use of a safe end shorter than five inches is not recommended, Fig. 11.

First Application.—New flue.

Second Application.—Cut swaged end to 3½ inches and apply 5 inches safe end, cutting to length after welding.

Third Application.—Cut swaged end to 3½ inches and apply safe end just long enough to give required length.

Fourth Application.—Cut off previous weld and reswage end of original dimensions. Apply safe end of full diameter at smokebox end, just long enough to give required length. Expand to standard requirements.

Fifth Application.—Same as second.

Sixth Application.—Same as third.

Seventh Application.—Same as fourth, except that old safe end at smokebox end of the flue must be cut off and a longer safe end used.

Suggestions in Connection With the Ordering, Unloading and Storing of the Tubes

Ordering.—Requisitions or orders for boiler tubes, arch tubes, superheater flues and safe ends, for new work or replacements, should clearly state:

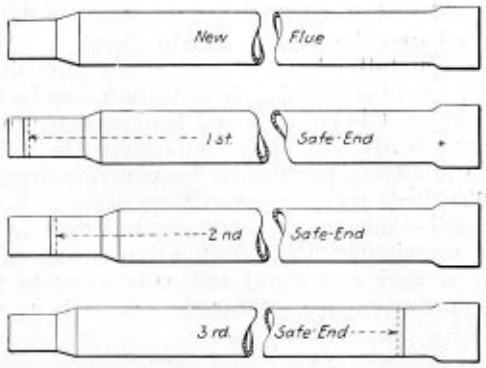


Fig. 11.—Successive Applications of Safe Ends

Length ..... in feet or inches.
Outside diameter ..... in inches
Thickness ..... in B. W. G. number, or in decimals of an inch.

The thickness of heavy wall arch tubes, water bars and marine stay tubes is often expressed in fractions of an inch: 3/16 inch, ¼ inch, etc.

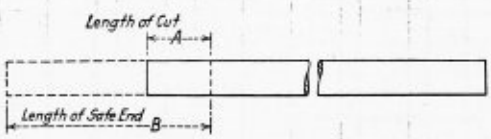


Fig. 12.—Lengths of Safe Ends

Table 2.—Application of Safe Ends

Application	Length of Cut A Inches	Length of End B Inches	Remarks
First	...	...	new flue
Second	1½	5	one weld
Third	5½	8	one weld
Fourth	8½	11	one weld
Fifth	11½	14	one weld
Sixth	1½	5	two welds
Seventh	5½	8	two welds
Eighth	8½	11	two welds
Ninth	27½	30	two welds
Tenth	1½	5	three welds

Table 3.—Decimal Equivalents of Birmingham Wire Gage (B.W.G.) Numbers Used in Boiler Tube Practice

Number	Thickness Birmingham Wire Gage Inches
13	0.095
12	0.109
11	0.120
10	0.134
9	0.148
8	0.165
7	0.180
6	0.203
5	0.220

If tubes or flues are to be swaged or expanded, the fact should be stated on the order, and a blueprint or sketch

PAR. 325 REVISED: Lugs or brackets, when used to support a boiler of any type, shall be properly fitted to the surfaces to which they are attached. The shearing and crushing stresses on the rivets used for attaching the lugs or brackets shall not exceed 8 percent of the strength given in paragraphs 15 and 16. WHERE IT IS IMPRACTICAL TO USE RIVETS, STUDS WITH NOT LESS THAN 10 THREADS PER INCH MAY BE USED. IN COMPUTING THE SHEARING STRESSES, THE AREA AT THE BOTTOM OF THE THREAD SHALL BE USED. [For traction or portable boilers, studs with pipe threads may be used.]

FIG. 32 REVISED:

A Change dimension line to center of stayrod.
D Change dimension line to center of staybolt.
F Change dimension line to center of door ring rivet.
H Give eighth sketch in drawing the designation "H."

PAR. 423.—CHANGE SECOND SECTION TO READ AS FOLLOWS:

A bevel-seated 3½ inch valve WITH [is marked by the manufacturer] 0.11 inch lift HAS A [and] discharge capacity AT [for] 100 pounds pressure OF [=] 4,840 pounds; hence two such valves would be required.

PAR. 424.—CHANGE SECOND SECTION TO READ AS FOLLOWS:

A bevel-seated 2½ inch valve WITH [is marked by the manufacturer] 0.08 inch lift HAS A [and] discharge capacity AT [for] 275 pounds pressure OF [=] 6,350 pounds; hence two such valves would be required.

PAR. 425.—CHANGE SECOND SECTION TO READ AS FOLLOWS:

A bevel-seated 2 inch valve WITH [is marked by the manufacturer] 0.07 inch lift HAS A [and] discharge capacity AT [for] 150 pounds pressure OF [=] 2,500 pounds; hence one such valve would be required.

TABLE 16.—CHANGE TITLE TO READ AS FOLLOWS:

AMERICAN STANDARD 125-POUNDS WORKING PRESSURE PER SQUARE INCH STANDARD CAST-IRON FLANGE FITTINGS, STRAIGHT SIZES (SEE FIG. 33).

TABLE 17.—CHANGE TITLE TO READ AS FOLLOWS:

AMERICAN STANDARD 250-POUNDS WORKING PRESSURE PER SQUARE INCH, EXTRA HEAVY CAST-IRON FLANGE FITTINGS, STRAIGHT SIZES (SEE FIG. 33).

PAR. 428 REVISED: Fusible plugs, if used, shall be filled with tin with a melting point between 400 and 500 degrees F., and shall be renewed once each year. WHERE THE BOILERS ARE TO BE OPERATED AT WORKING PRESSURES IN EXCESS OF 225 POUNDS PER SQUARE INCH GAGE, THE USE OF FUSIBLE PLUGS IS NOT ADVISABLE.

Fusible Plugs.

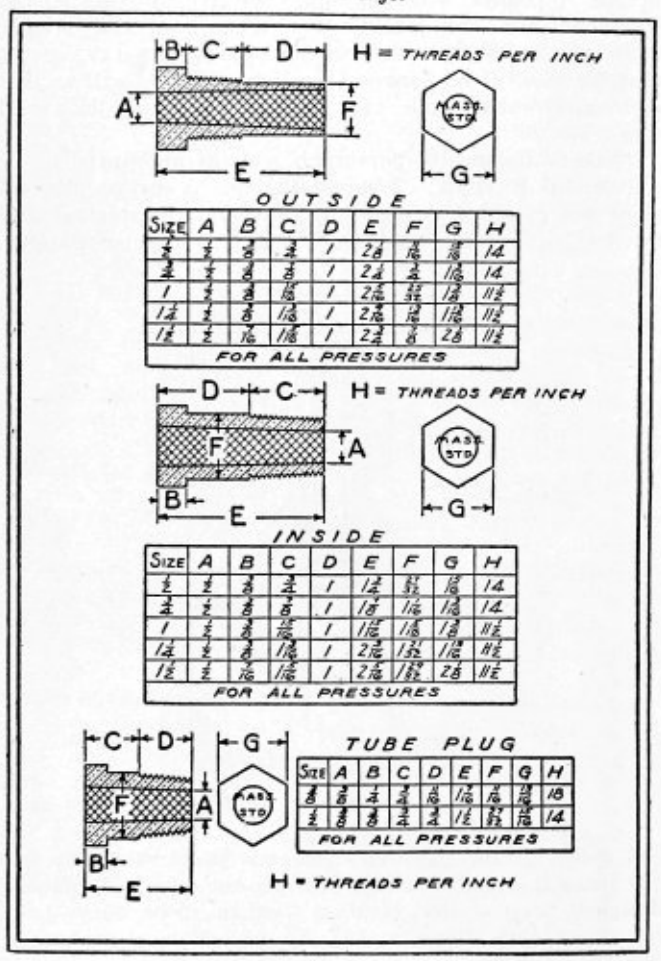


Fig. 33½.—Acceptable Forms of Fusible Plugs

Relative Advantages of Non-Lifting and Lifting Injectors*

FOR the purposes of securing data on this subject a questionnaire was prepared and sent to sixteen railroads from which eleven replies have been received. The advantages of the non-lifting injectors are reported as follows: (1) Larger capacity for same size injector, injector pipes and connections; (2) grades closer, giving a wider range for each size of injector; (3) gives less trouble from restricted openings in the feed water supply pipe; (4) located on the outside of the cab, thus relieving congested conditions inside; (5) more reliable and dependable; (6) application and maintenance cost less; (7) more accessible for repairs and repairs can be be made on the injector in place without disconnecting from the piping; (8) will work with less water in tank, also operate with feed water at a considerably higher temperature.

The disadvantages of this type injector are its greater liability to being knocked off by derailment, side swipe, etc., and the liability to freeze in extremely cold weather.

ADVANTAGES AND DISADVANTAGES OF INJECTORS

The advantages of the lifting injectors are stated as follows: (1) Mounted in cab where engineman can more easily see and observe its workings; (2) if trouble occurs on the road it is easier for the engineer to remedy it; (3) self contained and fewer parts to operate; (4) easier operated and controlled.

Its disadvantages are: (1) Less grading range than the non-lifter; (2) more readily affected by restricted feed water opening and length of lift; (3) must be taken down from pipe connections to repair the tubes; (4) tubes coat up more rapidly than in the non-lifters; (5) it is subject to failure from a leaky boiler check, worn steam nozzle or combining tube, leak in feed pipe, or if injectors become overheated they must be cooled off before they will operate.

It would appear from the replies received that:

(a) Little is known about the comparative efficiency of the two types of injectors with respect to steam consumption for a given quantity of water delivered; (b) the cost of repairs and maintenance is less with the non-lifting injector; (c) the non-lifting injector is more easily located and repaired; (d) the non-lifting injector is generally considered preferable and is gaining in favor, particularly on large locomotives.

*Section of locomotive construction committee report read at Atlantic City convention of the American Railway Association—Division V—Mechanical, held June 14, 1922.

What to Consider in Selecting Staybolts*

By C. A. Seley†

The subject of staybolts for steam boilers is not by any means a new one and so the following article is in the nature of a review of recent developments in staybolt design and production. Staybolts are necessary in many types of boilers for staying flat surfaces; in the Scotch marine for staying the combustion chambers; in watertube boilers having box headers and most important of all in the fireboxes of locomotive boilers. Differences in the material and design of staybolts for various classes of service must be considered and suggestions are given below that will aid in properly determining the types to be used.

NOT all boilers are afflicted with staybolts, if such a term may be borrowed from authorities of the United States Navy, whose officers do not favor designs of boilers that require staybolts, but prefer watertube types with tubes, headers and drums arranged to avoid the need of staybolts. Many boilers, however, do need staybolts and these will be discussed.

In the Scotch marine type boiler, staybolts are required for staying the flat surfaces of the furnace combustion chambers. Box headers of watertube type boilers are used extensively in land and sea service which require staybolts, but by far the largest number is used in the locomotive type boiler, due to the great number of flat surfaces to be stayed.

In our day there have been attempts to utilize sectional firebox construction which would avoid the use of staybolts, but apparently these have not been successful. Some years ago there was an attempt to utilize the corrugated type of furnace, common in marine practice, in locomotive boilers, but the disadvantages of limited firebox and grate areas were not as well understood in those days as in later years or the idea would never have been considered.

Materials for Staybolts

The quite general practice of using copper fireboxes and copper or bronze staybolts abroad has not been favored in this country where steel for sheets and iron for bolts is the general practice in locomotive boilers. On account of price and their wonderful properties, modern alloy steels, having extremes of strength and ductility, great vibratory performance in laboratory testing machines, certainly deserve consideration for staybolt work.

Under the United States marine boiler inspection rules, steel is permissible and is used to a considerable extent. One reason is that the service stresses in ordinary marine boilers are by no means as severe as in a locomotive. At the same time the old marine engineer will tell you that if he can replace steel staybolts with iron ones, the trouble ceases. Steel for staybolts for locomotives has never gone beyond the experimental stage in this country, to our knowledge, for very good and sufficient reasons. These reasons are mainly the manner of breakage. Due to the homogeneity of the material the fracture is not progressive. One bolt breaking is likely to set off others in the vicinity, thus breaking in bunches and creating a very unsafe condition, not as readily under control as is the case with iron staybolts, in which fractures are progressive.

The successful material judging from the general practice of builders and users of locomotives in this country is properly manufactured iron very adequately covered by specifications approved by the mechanical section of the American Railway Association, formerly the American Railway Master Mechanics' Association.

*Paper read at a meeting of the Pittsburgh Railway Club, November 18.
†President of the American Flexible Bolt Company, Pittsburgh, Pa.

Maintenance of Staybolts

One lesson that the mechanical man in any line has to learn is that if a thing breaks it should not necessarily be made bigger. It may be too big and fail on account of stiffness and rigidity. This is a conclusion well illustrated in the maintenance of staybolts. Due to the severity of service conditions, a locomotive boiler at one moment may be developing a thousand horsepower or over, followed by a sudden cessation of demand, causing a maximum variation of temperature in various parts of the firebox. Dumping of fires and rapid and uneven cooling down; washing and water changing; rapid firing up; extreme external temperature conditions, all contribute to relatively unequal movements of various parts of the firebox and combustion chamber with reference to the outer wrapper sheet and shell. It has been well stated that a locomotive firebox, except when cold, is a constant vibrating machine acting on the staybolts.

Here then is the element of service that deserves special consideration. It is easy enough to determine the spacing, pressure per square inch and the total static load on any staybolt and arrive at the proper dimensions to fulfill safety and statutory requirements. That work is done before the locomotive is built. After going into service it may not be many months or years before there is cause for staybolt renewal. It is so easy to run in the next larger size staybolt tap to renew the threads in the sheets that the staybolts grow rather rapidly sometimes and the short life and unsatisfactory service of over-grown staybolts have, on many roads now led to placing a limit, after which staybolt holes and sizes must be reduced by welding down, bushing or use of new sheets. This was and is particularly desirable with the so-called rigid staybolt, ordinarily threaded full length, which, when over-grown is a vicious lever to buckle, work, crack and retire the firebox sheets, causing a renewal at high cost and loss of serviceability. Before the firebox surrenders it may break a lot of staybolts and the bigger they are the quicker they break, adding to expense and loss of engine service.

Breakage of Rigid Bolts

It is well known and understood that 99 percent plus of the breakage of ordinary rigid staybolts occurs at the end close to the inner surface of outside sheet. In order to detect these breaks readily the outer ends are drilled with tell-tale holes, at least 1¼ inches deep, to comply with Interstate Commerce Commission boiler rules. This depth extends generally ¾ inch or well beyond the usual point of failure and usually announces a fractured or broken bolt by leakage in case it is not previously detected by hammer test.

In order not to have to tear down brick work in oil burners or arches every thirty days or in the case of staybolts located in places inaccessible for convenient inspection and hammer test, the Interstate Commerce Commission rules permit the use of staybolts with tell-tale holes throughout their length. This is a measure for convenience and safety but